

publish

Generate view of MATLAB file in specified format

Syntax

```
publish(file)
publish(file,format)

publish(file,Name,Value)
publish(file,options)

my_doc = publish(file, __ )
```

Description

`publish(file)` generates a view of the specified MATLAB[®] code file and output in an HTML format that can be used for sharing. `publish` saves the HTML file and a file for each graphic that the code creates in a subfolder named `html`. The location of the `html` subfolder is relative to the location of `file`. [example](#)

For example, `publish('C:\myMATLABfiles\myfile.m')` runs the code in `myfile.m` using the base workspace, and then saves the formatted code and results in `C:\myMATLABfiles\html\myfile.html`.

`publish(file,format)` generates a view of the specified MATLAB file in the specified file format. All file formats save to the `html` subfolder. [example](#)

`publish(file,Name,Value)` generates a view of the specified MATLAB file with options specified by one or more name, value pair arguments. [example](#)

`publish(file,options)` uses the options structure to generate the view of the specified MATLAB file. Using a structure to specify options is useful when you want to preconfigure and save your options for repeated use. The fields and values of the options structure correspond to names and values of name-value pair arguments. [example](#)

`my_doc = publish(file, __ )` generates a view of the specified MATLAB file and returns the path of the resulting output file. You can use this syntax with any of the input argument combinations in the previous syntaxes. [example](#)

Examples

[collapse all](#)

Generate HTML View of MATLAB Script

Create an HTML view of a MATLAB example script including the code, results, and comments. Use the `web` function to view the resulting file.

```
publish("fourier_demo2.m");
web("html/fourier_demo2.html")
```

Generate View of MATLAB Script in Microsoft Word Format

Create a Microsoft® Word view of a MATLAB example script including the code, results, and comments. Save the path of the published file to a variable. Use the `winopen` function to view the resulting file.

```
mydoc = publish("fourier_demo2.m","doc");  
winopen(mydoc)
```

Customize View of MATLAB Script Using Name-Value Arguments

Use name-value arguments to change the appearance of published figure windows in the HTML view of a MATLAB example script.

Use the `figureSnapMethod` name-value argument with the value `entireFigureWindow` to include window decorations and to match the figure background color to the screen color for figures. Then, use the `web` function to view the resulting file.

```
publish("fourier_demo2.m","figureSnapMethod","entireFigureWindow");  
web("html/fourier_demo2.html")
```

Customize View of MATLAB Script Using Structure

Use a structure to create a Microsoft Word view of a MATLAB example script that does not show the code from the script. Specifying options as a structure is useful when you want to preconfigure and save your options for repeated use.

Create a structure *options* that specifies the format and `showcode` options.

```
options.format = "doc";  
options.showCode = false;
```

Create a Microsoft Word view of the example using the options specified in *options*. Use the `winopen` function to view the resulting file.

```
publish("fourier_demo2.m",options);  
winopen("html/fourier_demo2.doc")
```

Generate HTML View of MATLAB Function with Input Arguments

Generate an HTML view of a MATLAB function that requires input arguments.

Create and save the function `fact.m`.

```
% FACT compute the factorial of a number  
% FACT(N) computes the factorial of the number N and returns the results  
function f = fact(n)  
    f = prod(1:n);  
end
```

Use the `publish` function to generate an HTML view of the function. Set the value of the input argument n by using the `codeToEvaluate` name-value argument. Then, use the `web` function to view the resulting file.

```
publish("fact.m", "codeToEvaluate", "fact(5);");
web("html/fact.html")
```

Input Arguments

collapse all

file — MATLAB file name
character vector | string

MATLAB file name, specified as a character vector or string. `file` can include a full or partial path.

 Note

When MATLAB publishes a file, it can overwrite existing files from the output folder that start with the same name as `file`.

Example: `publish('myfile.m')`

Example: `publish('C:\myMATLABfiles\myfile.m')`

format — Output format
'html' (default) | 'doc' | 'latex' | 'ppt' | 'xml' | 'pdf'

Output format of published file, specified as one of the values listed in the table.

Value	Output Format
'html' (default)	Hypertext Markup Language
'doc'	Microsoft Word
'latex'	LaTeX
'ppt'	Microsoft PowerPoint®
'xml'	Extensible Markup Language
'pdf'	Portable Document Format (PDF)

The Microsoft Word and Microsoft PowerPoint formats are only available on Windows® platforms.

MATLAB does not preserve syntax highlighting when you specify the output format as Microsoft PowerPoint or LaTeX.

Example: `publish('myfile.m', 'ppt');`

options — Options for published output

structure

Options for published output, specified as a structure. Use the `options` structure instead of name-value pair arguments when you want to reuse the same configuration for publishing multiple MATLAB files.

The fields and values of the `options` structure correspond to names and values of the name-value pair arguments.

For example, this command creates the structure `options`, and specifies the PDF output format and the output folder `C:\myPublishedOutput`.

```
options = struct('format','pdf','outputDir','C:\myPublishedOutput')
```

Name-Value Arguments

[expand all](#)

Specify optional pairs of arguments as `Name1=Value1, ..., NameN=ValueN`, where `Name` is the argument name and `Value` is the corresponding value. Name-value arguments must appear after other arguments, but the order of the pairs does not matter.

Before R2021a, use commas to separate each name and value, and enclose `Name` in quotes.

Example: `'format','pdf','showCode',false` specifies the PDF output file format and excludes the code from the output.

Output Options

[expand all](#)**format — Output format**

'html' (default) | 'doc' | 'latex' | 'ppt' | 'xml' | 'pdf'

outputDir — Output folder

character vector

stylesheet — Extensible Stylesheet Language (XSL) file

character vector

Figure Options

[expand all](#)**createThumbnail — Whether to create thumbnail image**

true (default) | false

figureSnapMethod — Figure window appearance

'entireGUIWindow' (default) | 'print' | 'getframe'

imageFormat — Image file format

'png' | 'eps' | 'jpeg' | ...

maxHeight — Maximum image height

[] (default) | positive integer

maxWidth — Maximum image width

[] (default) | positive integer

useNewFigure — Whether to create new figure

true (default) | false

Code Options

[expand all](#)**evalCode — Whether to run code**

true (default) | false

catchError — Whether to catch errors

true (default) | false

codeToEvaluate — Additional code to run

character vector

maxOutputLines — Maximum number of lines

Inf (default) | nonnegative integer value

showCode — Whether to include code

true (default) | false

Limitations

- Publishing live scripts or functions is not supported. To convert live scripts or functions to other formats, use the [export](#) function instead.
- Publishing a MATLAB code file (.m) by calling the publish function from a live script or function (.mlx) is not supported.
- The publish function does not support the input function. Publishing a MATLAB code file (.m) that contains a call to the input function results in an error.

Tips

- To enhance the readability of the published document and include additional image snapshots, external file content, and external images, see [Publishing Markup](#).
- To avoid duplicate snapshots when publishing MATLAB code files, make sure that your cursor is not positioned over any figures before calling the publish function. Having the cursor on a figure can cause the figure to update and the publish function to generate duplicate snapshots.

Version History

Introduced before R2006a

R2025a: "entireFigureWindow" value no longer supported for figureSnapMethod name-value argument ⚠

R2025a: "bmp" value no longer supported for imageFormat name-value argument with "pdf" format or "print" figureSnapMethod name-value argument ⚠

R2025a: Figures appear larger and as tabs by default in published documents ⚠

R2024b: 'entireFigureWindow' value for figureSnapMethod option will not be supported ⚠

See Also

[grabcode](#) | [snapnow](#) | [web](#) | [winopen](#) | [export](#)

Topics

[Publish and Share MATLAB Code](#)

[Publishing Markup](#)

[Ways to Share and Export Live Scripts and Functions](#)